

Nikhil Kumar Reddy Pyreddy

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SUMMARY

Full-stack software engineer with five years of hands-on engineering craftsmanship across multiple programming languages and modern frameworks, delivering high-quality, outcome-focused solutions on cloud-native architectures. Proficient in C#/.NET 8, Java/Spring Boot, Python, Node.js, TypeScript/JavaScript, React, and Angular, with strong SQL/NoSQL across MS SQL Server, MySQL, PostgreSQL, and MongoDB. At Paycom, designs and ships end-to-end products spanning user interfaces, middle-tier services, and data layers; integrates GenAI and LLM-driven orchestration into production workflows powering AI-driven search and developer tooling on Apache Kafka event streams. Embraces DevSecOps, SRE, and lean, iterative delivery using SAFe, GitHub, SonarQube, and full CI/CD automation through SDLC. Strong foundation in OOP, data structures, algorithms, and modern coding practices.

TECHNICAL SKILLS

Languages: C#, Java, Python, TypeScript, JavaScript (ES6+), SQL, HTML5, CSS3/SASS

Frameworks & APIs: .NET 8, ASP.NET Core (Web API, MVC), Spring Boot, Spring MVC, Hibernate/JPA, Node.js, Flask, React, Angular, REST, GraphQL, RPC

Cloud-Native (FaaS / PaaS / Micro-services): AWS (Lambda, S3, CloudWatch, API Gateway), GCP (Cloud Essentials certified), Docker, Kubernetes, HashiCorp Vault, serverless architectures, micro-service design

Data Access & Databases: Dapper, Hibernate/JPA, ADO.NET; MS SQL Server (T-SQL), MySQL, PostgreSQL, MongoDB, SQLite; query optimization, schema design, stored procedures, indexing, connection pooling

Streaming, Testing & Quality: Apache Kafka (producer/consumer), event-driven architectures, ETL pipelines; xUnit, Moq, JUnit, AutoMapper, SonarQube, TDD, code reviews, code coverage

DevSecOps & Observability: GitHub, GitHub Actions, Azure DevOps, Maven, JIRA, Confluence, NLog, AWS CloudWatch, structured logging, monitoring, alerting, SRE practices, on-call support

AI / GenAI & Methodologies: LLM APIs (OpenAI, Claude), GitHub Copilot, LangChain, prompt engineering, AI agent orchestration, MCP (Model Context Protocol) integrations, retrieval-augmented generation; Agile, SAFe, XP, Lean, OOP, data structures, algorithms, design patterns

PROFESSIONAL EXPERIENCE

Software Developer | *Paycom Software, Inc.* | Grapevine, TX

Jun 2024 to Present

- Designed, developed, and shipped scalable end-to-end full-stack solutions in C#/.NET 8, React, and TypeScript for the Knowledge Base and Advanced Report Writer platforms, spanning user interfaces, middle-tier services, and data layers serving 36,000+ enterprise clients.
- Built data-access layers using Dapper with parameterized queries, stored procedures, AutoMapper-based entity-to-DTO mapping, and shared connection-management abstractions across SQL Server services, reducing data-access boilerplate by 40% while maintaining strong test coverage with xUnit and Moq enforced by SonarQube quality gates.
- Designed and rolled out caching layer optimizations across Knowledge Base Help Center APIs, reducing redundant database fetches by 55% and improving p95 response times by 45% for high-traffic article retrieval endpoints.
- Built cloud-native micro-services and serverless functions on Kubernetes, Docker, AWS Lambda, and HashiCorp Vault, deployed through fully automated CI/CD pipelines in GitHub Actions with staged deployments, structured NLog logging, and AWS CloudWatch monitoring and alerting.
- Engineered AI-augmented product features using LLM APIs (OpenAI, Claude) and prompt engineering, including LLM-driven orchestration that synchronizes Knowledge Base content with Data Science services to power generative AI search experiences.
- Built and integrated MCP (Model Context Protocol) and tool-orchestration layers that expose internal Knowledge Base, ARW, and developer-tooling capabilities to AI agents and Copilot-style assistants for retrieval, summarization, and workflow automation.
- Built event-driven streaming pipelines with Apache Kafka, designing producer and consumer services for near real-time data consistency between Knowledge Base content services and Data Science consumers.
- Contributed to the Confluence-to-Payges Knowledge Base platform migration covering 6,000+ articles, building API integration adapters, validation utilities, and automated migration scripts that achieved 97% automated validation accuracy and reduced manual review effort by 75%.
- Authored technical specifications, system designs, and test plans; collaborated cross-functionally with Product, QA, UX, and Data Science teams across two-week Agile/SAFe sprints to translate business needs into technical solutions.

Software Developer, Graduate Research | *University of Oklahoma* | Norman, OK

Aug 2022 to May 2024

- Owned the full lifecycle of two industry-sponsored engineering software products through rapid, iterative delivery, from architecture and prototyping to deployment, validation, and ongoing maintenance.

- Designed and developed a parametric simulation application from scratch, defining the calculation architecture and data model in collaboration with domain engineers, implementing core logic in Python, and shipping an analysis tool supporting 50+ configurable input variables.
- Built reusable ETL and data-processing pipelines with Python, NumPy, Pandas, and Matplotlib, ingesting and transforming simulation output datasets, applying schema transformation and data normalization, and producing automated reports and analytics dashboards.
- Authored software specifications, test plans, and test procedures validated against engineer-specified reliability requirements; iterated on algorithms based on validation feedback through weekly stakeholder syncs.
- Replaced manual spreadsheet workflows with documented, tested, reusable software components adopted into engineers' internal design workflows.

Software Development Intern | *Paycom Software, Inc.* | Dallas, TX

May 2023 to Aug 2023

- Designed and developed scalable web applications in C#/.NET 8 with ASP.NET Core MVC for enterprise customer data import workflows, reducing data quality issues by 60%.
- Built front-end interfaces in React and ASP.NET Core MVC with REST API integration, integrated SQL Server through stored procedures and parameterized queries, and added comprehensive unit and integration tests with automated coverage reporting.

Java Full Stack Developer | *Cognizant Technology Solutions* (Client: AIG UK) | Hyderabad, India

Feb 2021 to Jul 2022

- Developed a commercial policy lifecycle application for AIG UK (Fortune 500 global insurer) using Java, J2EE, Spring Boot, Spring MVC, and Hibernate/JPA, building RESTful Web APIs and back-end micro-services for end-to-end commercial policy management workflows on SQL Server with Maven-based builds.
- Designed and implemented Spring Boot REST API integrations with third-party insurance systems, building customer-facing portals and internal enterprise tools using Angular, React, HTML5, Bootstrap, and CSS3 across distributed teams.
- Built Apache Kafka producer and consumer services for event-driven messaging across the policy platform, supporting asynchronous workflows and decoupled communication between policy lifecycle micro-services.
- Performed database query tuning, applied caching strategies, and optimized front-end bundles to reduce page load times by 30%; integrated AWS S3 with Azure DevOps CI/CD pipelines featuring JUnit-based automated testing and staged deployments.
- Collaborated with Business Analysts and stakeholders to translate insurance domain requirements into technical specifications and accurate implementation of complex policy business rules across a 30 to 40 engineer distributed Java/Spring Boot program spanning India and UK.

EDUCATION

Master of Science in Computer Science

Aug 2022 to May 2024

University of Oklahoma, Norman, OK • GPA: 3.7/4.0

Coursework: Database Systems; Parallel, Distributed & Network Programming; Algorithm Analysis; Software Engineering Processes; Computer Architecture; Computer Security; Machine Learning.

Bachelor of Technology in Computer Science & Engineering

Jul 2017 to Jun 2021

Vardhaman College of Engineering, Hyderabad, India • GPA: 8.12/10

PROJECTS

High-Performance AI-Augmented Web Crawler | Python, AsyncIO, LangChain, LLM APIs, Redis, BeautifulSoup

- Designed a concurrent crawling system using thread pools and asynchronous I/O to scrape and process thousands of pages per minute with adaptive rate limiting and retry handling.
- Integrated LLM-based content classification and structured data extraction using LangChain and OpenAI APIs, replacing brittle regex and CSS-selector parsing with adaptive AI-driven extraction resilient to page-structure changes.
- Implemented URL deduplication, distributed task queues, and fault-tolerant checkpointing using Redis for scalable parallel execution.

Norman Police Department Data Redaction Tool | Python, NLP, scikit-learn, spaCy, Regex, SQLite

- Built a back-end redaction tool that identifies and conceals sensitive PII in police department documents using a Named Entity Recognition pipeline (spaCy and scikit-learn) combined with regex pattern matching, NLP text vectorization, and SQLite for data handling.

PUBLICATIONS & CERTIFICATIONS

Publications: *Image Dehazing for Surveillance Systems*, 4th Smart Cities Symposium (SCS 2021), IET Digital Library [\[Link\]](#); *URL Shortening Service Using Flask & Base62 Algorithm*, International Conference on Intelligent Computing and Networking (ICICN 2021).

Certifications: Google Cloud Essentials • AWS Academy Cloud Foundations • MongoDB Basics • Python (HackerRank) • Cisco Cybersecurity Essentials • Responsive Web Design (freeCodeCamp).